

## ПУНКТ 1

Для изоляции трафика на канальном уровне (L2) были созданы виртуальные локальные сети VLAN 20 и VLAN 333. Для межкоммутаторных магистральных портов (Trunk) настроен Native VLAN 333, что позволяет передавать кадры этого VLAN без тегирования

На коммутаторах доступа порты до компьютеров переведены в режим доступа (access), а все четыре восходящих канала к коммутаторам ядра объединены в транки с Native VLAN 333

### Конфигурация коммутатора Layer2Switch-3:

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 20
Switch(config-vlan)#name VLAN_20
Switch(config-vlan)#vlan 333
Switch(config-vlan)#name VLAN_333
Switch(config-vlan)#exit
Switch(config)#interface range GigabitEthernet0/0 - 3
Switch(config-if-range)#switchport trunk encapsulation dot1q
Switch(config-if-range)#switchport mode trunk
Switch(config-if-range)#switchport trunk native vlan 333
Switch(config-if-range)#exit
Switch(config)#interface range GigabitEthernet1/0 - 3
Switch(config-if-range)#switchport trunk encapsulation dot1q
Switch(config-if-range)#switchport mode trunk
Switch(config-if-range)#switchport trunk native vlan 333
Switch(config-if-range)#exit
Switch(config)#end
Switch#write memory
Building configuration...
```

### Конфигурация коммутатора Layer2Switch-4:

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 20
Switch(config-vlan)#name VLAN_20
```

```
Switch(config-vlan)#vlan 333
Switch(config-vlan)#name VLAN_333
Switch(config-vlan)#exit
Switch(config)#interface range GigabitEthernet0/0 - 3
Switch(config-if-range)#switchport trunk encapsulation dot1q
Switch(config-if-range)#switchport mode trunk
Switch(config-if-range)#switchport trunk native vlan 333
Switch(config-if-range)#exit
Switch(config)#interface range GigabitEthernet1/0 - 3
Switch(config-if-range)#switchport trunk encapsulation dot1q
Switch(config-if-range)#switchport mode trunk
Switch(config-if-range)#switchport trunk native vlan 333
Switch(config-if-range)#exit
Switch(config)#end
Switch#write memory
Building configuration...
```

### Конфигурация коммутатора Layer2Switch-5:

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 20
Switch(config-vlan)#name VLAN_20
Switch(config-vlan)#vlan 333
Switch(config-vlan)#name VLAAN_333
Switch(config-vlan)#exit
Switch(config)#interface GigabitEthernet1/0
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on GigabitEthernet1/0 but will only
have effect when the interface is in a non-trunking mode.
Switch(config-if)#exit
Switch(config)#interface GigabitEthernet1/1
Switch(config-if)#switchport mode access
```

```
Switch(config-if)#switchport access vlan 333
Switch(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on GigabitEthernet1/1 but will only
have effect when the interface is in a non-trunking mode.
Switch(config-if)#exit
Switch(config)#interface range GigabitEthernet0/0 - 3
Switch(config-if-range)#switchport trunk encapsulation dot1q
Switch(config-if-range)#switchport mode trunk
Switch(config-if-range)#switchport trunk native vlan 333
Switch(config-if-range)#exit
Switch(config)#end
Switch#write memory
Building configuration...
```

На коммутаторах ядра создаются аналогичные VLAN, а все активные межкоммутаторные порты (с Ethernet0/0 по Ethernet1/3) переводятся в режим транка с Native VLAN 333

### **Конфигурация коммутатора Layer2Switch-1:**

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 20
Switch(config-vlan)#name VLAN_20
Switch(config-vlan)#vlan 333
Switch(config-vlan)#name VLAN_333
Switch(config-vlan)#exit

Switch(config)#interface GigabitEthernet1/0
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
```

Use with CAUTION

%Portfast has been configured on GigabitEthernet1/0 but will only have effect when the interface is in a non-trunking mode.

Switch(config-if)#exit

Switch(config)#interface GigabitEthernet1/1

Switch(config-if)#switchport mode access

Switch(config-if)#switchport access vlan 333

Switch(config-if)#spanning-tree portfast

%Warning: portfast should only be enabled on ports connected to a single host. Connecting hubs, concentrators, switches, bridges, etc... to this interface when portfast is enabled, can cause temporary bridging loops.

Use with CAUTION

%Portfast has been configured on GigabitEthernet1/1 but will only have effect when the interface is in a non-trunking mode.

Switch(config-if)#exit

Switch(config)#interface range GigabitEthernet0/0 - 3

Switch(config-if-range)#switchport trunk encapsulation dot1q

Switch(config-if-range)#switchport mode trunk

Switch(config-if-range)#switchport trunk native vlan 333

Switch(config-if-range)#exit

Switch(config)#end

Switch#write memory

Building configuration...

## Конфигурация коммутатора Layer2Switch-2:

Switch>enable

Switch#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Switch(config)#vlan 20

Switch(config-vlan)#name VLAN\_20

Switch(config-vlan)#vlan 333

Switch(config-vlan)#name VLAN\_333

Switch(config-vlan)#exit

Switch(config)#interface GigabitEthernet1/0

Switch(config-if)#switchport mode access

Switch(config-if)#switchport access vlan 20

Switch(config-if)#spanning-tree portfast

%Warning: portfast should only be enabled on ports connected to a single

host. Connecting hubs, concentrators, switches, bridges, etc... to this interface when portfast is enabled, can cause temporary bridging loops.  
Use with CAUTION

%Portfast has been configured on GigabitEthernet1/0 but will only have effect when the interface is in a non-trunking mode.

```
Switch(config-if)#exit
```

```
Switch(config)#interface GigabitEthernet1/1
```

```
Switch(config-if)#switchport mode access
```

```
Switch(config-if)#switchport access vlan 333
```

```
Switch(config-if)#spanning-tree portfast
```

%Warning: portfast should only be enabled on ports connected to a single host. Connecting hubs, concentrators, switches, bridges, etc... to this interface when portfast is enabled, can cause temporary bridging loops.  
Use with CAUTION

%Portfast has been configured on GigabitEthernet1/1 but will only have effect when the interface is in a non-trunking mode.

```
Switch(config-vlan)#exit
```

```
Switch(config)#interface range GigabitEthernet0/0 - 3
```

```
Switch(config-if-range)#switchport trunk enca
```

```
Switch(config-if-range)#switchport trunk encapsulation dot
```

```
Switch(config-if-range)#switchport trunk encapsulation dot1q
```

```
Switch(config-if-range)#switchport mode trunk
```

```
Switch(config-if-range)#switchport trunk native vlan 333
```

```
Switch(config-if-range)#exit
```

```
Switch(config)#end
```

```
Switch#write memory
```

```
Building configuration...
```

## ПУНКТ 2

Для проверки изоляции и доступности хостов на персональных компьютерах настроены IP-адреса из соответствующих подсетей:

### **VLAN 20 (Подсеть 192.168.20.0/24):**

PC1> ip 192.168.20.1 255.255.255.0

PC3> ip 192.168.20.3 255.255.255.0

PC5> ip 192.168.20.5 255.255.255.0

### **VLAN 333 (Подсеть 192.168.33.0/24):**

PC2> ip 192.168.33.2 255.255.255.0

PC4> ip 192.168.33.4 255.255.255.0

PC6> ip 192.168.33.6 255.255.255.0

### **Результаты тестирования:**

- **Внутри VLAN 20:** Пинг с PC1 на PC3 (ping 192.168.20.3) проходит успешно.
- **Внутри VLAN 333:** Пинг с PC2 на PC4 (ping 192.168.33.4) проходит успешно.
- **Между VLAN 20 и VLAN 333:** Пинг с PC1 на PC2 (ping 192.168.33.2) неуспешен (хосты изолированы на канальном уровне).

### ПУНКТ 3

При отправке пакетов зафиксировано различное поведение заголовков Ethernet-кадров:

#### Анализ трафика VLAN 20 (Тегированный):

При передаче ICMP-пакета от PC1 к PC3 в заголовке Ethernet II присутствует дополнительное поле 802.1Q Virtual LAN

Данные заголовка: Type: 802.1Q Virtual LAN (0x8100); ID: 20

#### Анализ трафика VLAN 333 (Нетегированный / Native):

При передаче ICMP-пакета от PC2 к PC4 поле 802.1Q Virtual LAN в структуре кадра полностью отсутствует. Кадр передается в исходном виде (от MAC-адресов сразу к IPv4). Поскольку VLAN 333 назначен в качестве Native VLAN на транках, коммутаторы передают его кадры без тегов, что подтверждает соответствие стандарту IEEE 802.1Q

Layer2Switch-3\_Ethernet2\_to\_Layer2Switch-2\_Ethernet2.pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

(arp || icmp)

| No. | Time      | Source           | Destination      | Protocol | Length | Info                                                            |
|-----|-----------|------------------|------------------|----------|--------|-----------------------------------------------------------------|
| 15  | 10.379487 | Private_66:68:00 | Broadcast        | ARP      | 68     | Who has 192.168.20.3? Tell 192.168.20.1                         |
| 16  | 10.433937 | Private_66:68:02 | Private_66:68:00 | ARP      | 68     | 192.168.20.3 is at 00:50:79:66:68:02                            |
| 17  | 10.440714 | 192.168.20.1     | 192.168.20.3     | ICMP     | 102    | Echo (ping) request id=0x6aef, seq=1/256, ttl=64 (reply in 18)  |
| 18  | 10.460361 | 192.168.20.3     | 192.168.20.1     | ICMP     | 102    | Echo (ping) reply id=0x6aef, seq=1/256, ttl=64 (request in 17)  |
| 26  | 11.468473 | 192.168.20.1     | 192.168.20.3     | ICMP     | 102    | Echo (ping) request id=0x6bef, seq=2/512, ttl=64 (reply in 27)  |
| 27  | 11.506002 | 192.168.20.3     | 192.168.20.1     | ICMP     | 102    | Echo (ping) reply id=0x6bef, seq=2/512, ttl=64 (request in 26)  |
| 28  | 12.567695 | 192.168.20.1     | 192.168.20.3     | ICMP     | 102    | Echo (ping) request id=0x6cef, seq=3/768, ttl=64 (reply in 29)  |
| 29  | 12.626217 | 192.168.20.3     | 192.168.20.1     | ICMP     | 102    | Echo (ping) reply id=0x6cef, seq=3/768, ttl=64 (request in 28)  |
| 30  | 13.645640 | 192.168.20.1     | 192.168.20.3     | ICMP     | 102    | Echo (ping) request id=0x6eef, seq=4/1024, ttl=64 (reply in 31) |
| 31  | 13.661285 | 192.168.20.3     | 192.168.20.1     | ICMP     | 102    | Echo (ping) reply id=0x6eef, seq=4/1024, ttl=64 (request in 30) |
| 32  | 14.685172 | 192.168.20.1     | 192.168.20.3     | ICMP     | 102    | Echo (ping) request id=0x6fef, seq=5/1280, ttl=64 (reply in 33) |
| 33  | 14.716911 | 192.168.20.3     | 192.168.20.1     | ICMP     | 102    | Echo (ping) reply id=0x6fef, seq=5/1280, ttl=64 (request in 32) |
| 64  | 31.706092 | Private_66:68:01 | Broadcast        | ARP      | 64     | Who has 192.168.33.4? Tell 192.168.33.2                         |
| 65  | 31.730322 | Private_66:68:03 | Private_66:68:01 | ARP      | 64     | 192.168.33.4 is at 00:50:79:66:68:03                            |
| 66  | 31.743006 | 192.168.33.2     | 192.168.33.4     | ICMP     | 98     | Echo (ping) request id=0x80ef, seq=1/256, ttl=64 (reply in 67)  |
| 67  | 31.759593 | 192.168.33.4     | 192.168.33.2     | ICMP     | 98     | Echo (ping) reply id=0x80ef, seq=1/256, ttl=64 (request in 66)  |
| 68  | 32.769986 | 192.168.33.2     | 192.168.33.4     | ICMP     | 98     | Echo (ping) request id=0x81ef, seq=2/512, ttl=64 (reply in 69)  |
| 69  | 32.848888 | 192.168.33.4     | 192.168.33.2     | ICMP     | 98     | Echo (ping) reply id=0x81ef, seq=2/512, ttl=64 (request in 68)  |
| 70  | 33.860963 | 192.168.33.2     | 192.168.33.4     | ICMP     | 98     | Echo (ping) request id=0x82ef, seq=3/768, ttl=64 (reply in 71)  |
| 71  | 33.882872 | 192.168.33.4     | 192.168.33.2     | ICMP     | 98     | Echo (ping) reply id=0x82ef, seq=3/768, ttl=64 (request in 70)  |
| 72  | 34.905947 | 192.168.33.2     | 192.168.33.4     | ICMP     | 98     | Echo (ping) request id=0x83ef, seq=4/1024, ttl=64 (reply in 73) |
| 73  | 34.959052 | 192.168.33.4     | 192.168.33.2     | ICMP     | 98     | Echo (ping) reply id=0x83ef, seq=4/1024, ttl=64 (request in 72) |
| 81  | 35.974733 | 192.168.33.2     | 192.168.33.4     | ICMP     | 98     | Echo (ping) request id=0x84ef, seq=5/1280, ttl=64 (reply in 82) |
| 82  | 36.016751 | 192.168.33.4     | 192.168.33.2     | ICMP     | 98     | Echo (ping) reply id=0x84ef, seq=5/1280, ttl=64 (request in 81) |

Frame 17: Packet, 102 bytes on wire (816 bits), 102 bytes captured (816 bits) on interface -, id 0

Ethernet II, Src: Private\_66:68:00 (00:50:79:66:68:00), Dst: Private\_66:68:02 (00:50:79:66:68:02)

802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 20

000. .... = Priority: Best Effort (default) (0)

...0 .... = DEI: Ineligible

.... 0000 0001 0100 = ID: 20

Type: IPv4 (0x0800)

Internet Protocol Version 4, Src: 192.168.20.1, Dst: 192.168.20.3

Internet Control Message Protocol

Frame 66: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0

Ethernet II, Src: Private\_66:68:01 (00:50:79:66:68:01), Dst: Private\_66:68:03 (00:50:79:66:68:03)

Internet Protocol Version 4, Src: 192.168.33.2, Dst: 192.168.33.4

Internet Control Message Protocol

#### ПУНКТ 4

Все конфигурации сохранены в соответствующих файлах Switch1/2/3/4/5-config.txt

## ПУНКТ 5

Для организации связи между VLAN 20 и VLAN 333 в схему добавлен маршрутизатор R1

- Линк R1 <-> Layer2Switch-1 настраиваем в **VLAN 20**.
- Линк R1 <-> Layer2Switch-2 настраиваем в **VLAN 333**.

### 1. Настройка коммутатора Layer2Switch-1

Переводим порт подключения роутера в режим доступа для VLAN 20:

```
Switch# configure terminal
interface Ethernet2/0
switchport mode access
switchport access vlan 20
spanning-tree portfast
end
write memory
```

### 2. Настройка коммутатора Layer2Switch-2

Переводим порт подключения роутера в режим доступа для VLAN 333:

```
Switch# configure terminal
interface Ethernet2/0
switchport mode access
switchport access vlan 333
spanning-tree portfast
end
write memory
```

### Настройка маршрутизатора R1

```
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface FastEthernet0/0
R1(config-if)#ip address 192.168.20.254 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#interface FastEthernet1/0
R1(config-if)#ip address 192.168.33.254 255.255.255.0
```

```
R1(config-if)#no shutdown
R1(config-if)#exit
*Mar 1 00:11:08.743: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed
state to up
*Mar 1 00:11:09.743: %LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet1/0, changed state to up
R1(config)#exit
R1#end
R1#write memory
Building configuration...
[OK]
```

## Проверка

```
PC1> ping 192.168.33.2

84 bytes from 192.168.33.2 icmp_seq=1 ttl=63 time=163.012 ms
84 bytes from 192.168.33.2 icmp_seq=2 ttl=63 time=94.357 ms
84 bytes from 192.168.33.2 icmp_seq=3 ttl=63 time=81.599 ms
84 bytes from 192.168.33.2 icmp_seq=4 ttl=63 time=151.075 ms
84 bytes from 192.168.33.2 icmp_seq=5 ttl=63 time=94.929 ms

PC1>
```